Covering systems with moduli of the form $p - 1$, $p \geq 5$ prime

Erdős Problem #273

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Abstract

A *covering system* of $\mathbb{Z}$ is a finite collection of arithmetic progressions whose union is the integers. Erdős Problem #273 ([1], p. 24) asks whether there exists a covering system of $\mathbb{Z}$ all of whose moduli are of the form $p-1$ for some prime $p \geq 5$ – equivalently, whose moduli lie in $\mathcal{M} = \{4, 6, 10, 12, 16, 18, 22, 28, 30, 36, 40, 42, 46, 52, 58, 60, \ldots\}$. We prove a density obstruction: any strict covering by progressions with moduli dividing a common period L must satisfy $\sum_m 1/m \geq 1$, the sum being taken over the distinct moduli used. Specialising to $L = 27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$, the set of admissible divisors $M_{27720} = \mathcal{M} \cap \operatorname{div}(27720)$ has 34 elements and satisfies $D(27720) := \sum_{m \in M_{27720}} 1/m = 953/990 < 1$, ruling out any strict admissible covering whose moduli all divide 27720; a refinement at $L = 83160 = 2^3 \cdot 3^3 \cdot 5 \cdot 7 \cdot 11$ strictly strengthens this by adjoining eleven further admissible moduli, and an independent bound at $L = 180180 = 2^2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ rules out covers built from a different family of moduli. The argument and its numerical specialisations are formalised in Lean 4 against Mathlib; `#print axioms` on each theorem reports only the three Lean foundational axioms.

1 Introduction

A *covering system* of the integers is a finite collection of arithmetic progressions

$$\{a_i \pmod{m_i}\}_{i=1}^{k}, \qquad m_i \geq 2,$$

whose union is $\mathbb{Z}$; equivalently, every integer lies in at least one residue class $a_i \pmod{m_i}$. Throughout this manuscript we adopt the Erdős–Graham convention that the moduli $m_1, \ldots, m_k$ are *distinct*; the resulting object is sometimes called a *strict* covering system in the recent literature (cf. the type `StrictCoveringSystem` in the DeepMind formal-conjectures Lean repository [5]). Without the distinctness constraint, every problem of the form "does there exist a covering of $\mathbb{Z}$ with moduli in a fixed set S containing some m?" trivialises, as $S \ni m$ alone admits the covering $\{0, 1, \ldots, m-1\} \pmod{m}$.

Covering systems were introduced by Erdős in 1950, and a sequence of problems concerning their structure have driven a substantial body of work in combinatorial number theory. The present manuscript concerns one of these problems, recorded as Problem #273 in the Erdős–Graham survey [1].

Conjecture 1 (Erdős #273)**.** *Does there exist a strict covering system $\{a_i \pmod{m_i}\}_{i=1}^{k}$ of $\mathbb{Z}$ such that each modulus m_i has the form $m_i = p_i - 1$ for some prime $p_i \geq 5$?*

Write $\mathcal{M} := \{p-1 : p \text{ prime}, \ p \geq 5\} = \{4, 6, 10, 12, 16, 18, 22, 28, 30, 36, 40, 42, 46, 52, 58, 60, \ldots\}$ for the set of allowed moduli. The question is whether $\mathbb{Z}$ admits a finite strict covering whose moduli are drawn from $\mathcal{M}$.

The restriction $p \geq 5$ is essential. Selfridge observed that allowing $p = 3$ admits a covering system using divisors of $360 = \operatorname{lcm}(2, 3, 4, 5, 6, 8, 9, 10, 12)$, all of which are of the form $p - 1$ for $p \in \{3, 5, 7\}$. The Selfridge construction crucially uses the modulus $2 = 3 - 1$. Forbidding

$p = 3$ eliminates the modulus 2 entirely and changes the problem: any covering by progressions modulo elements of $\mathcal{M}$ must account for every parity class without ever using the all-purpose modulus 2, and the second-smallest available modulus is 4.

Two pieces of recent machinery bear on the question. Hough [2] resolved the longstanding Erdős minimum-modulus problem by establishing a finite upper bound on the smallest modulus required in any covering system, ruling out covering systems with all moduli larger than a fixed constant. Balister, Bollobás, Morris and Sahasrabudhe [3] extended Hough's distortion method to a general non-existence framework for covering systems whose moduli are constrained to a multiplicatively structured set, lowering Hough's universal minimum-modulus bound from 10^{16} to 616,000. Neither result has been adapted to the present restriction in the published literature.

Conjecture 1 appears to be open as of 2026; no partial result—a lower or upper bound on the smallest required modulus, on the number of progressions, or on the least common multiple of any putative covering—appears in the published literature.

Main result. We establish an elementary density obstruction to bounded-period strict admissible coverings. The first ingredient is a counting union bound (Theorem 4) saying that any strict covering of $\mathbb{Z}$ by progressions whose moduli divide a common period L requires the sum of reciprocal moduli to be at least 1. Specialising to $L = 27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$, the set of admissible moduli dividing 27720 has 34 elements and the corresponding reciprocal-sum equals $953/990 < 1$, immediately ruling out any strict covering of $\mathbb{Z}$ by admissible progressions whose moduli all divide 27720 (Theorem 8).

This conclusion does not refute Conjecture 1: a putative covering may use moduli with LCM exceeding 27720. It is a lower bound on the structural complexity any solution must have. Equivalently: any covering of $\mathbb{Z}$ by progressions with moduli in $\mathcal{M}$ must use at least one modulus that does not divide 27720. Bounded-period SAT enumeration confirms the same non-existence at every L in $\{60, 120, 360, 840, 2520, 27720\}$ independently (Remark 12; Appendix B).

2 The density obstruction

Throughout this section, a *residue class* is a pair $(a, m) \in \mathbb{Z} \times \mathbb{N}_{\geq 2}$ together with the set $a + m\mathbb{Z} \subseteq \mathbb{Z}$.

Lemma 2. *Let (a, m) be a residue class with $m \mid L$, $L \geq 1$. Then the set $\{i \in [0, L) : i \equiv a \pmod{m}\}$ has cardinality exactly L/m.*

Proof. Write $L = mq$ with $q = L/m \in \mathbb{N}$. The map $j \mapsto a + mj \pmod{L}$ is a bijection between $\{0, 1, \ldots, q-1\}$ and $\{i \in [0, L) : i \equiv a \pmod{m}\}$. The lemma states the cardinality of the latter equals $q = L/m$. □

Lemma 3 (Density union bound)**.** *Let $L \geq 1$ and let S be a finite collection of residue classes (a_i, m_i) with $m_i \mid L$ for all i. Then*

$$\left|\{i \in [0, L) : \exists\, (a, m) \in S,\ i \equiv a \pmod{m}\}\right| \ \leq \sum_{(a,m) \in S} \frac{L}{m}.$$

Proof. Each residue class $(a, m) \in S$ covers a subset of $[0, L)$ of cardinality L/m by Lemma 2. The left-hand side is the cardinality of the union of these subsets, and the cardinality of a finite union is at most the sum of the cardinalities (the union bound). □

The main density theorem is the contrapositive of Lemma 3 together with the strictness assumption.

Theorem 4 (Strict-covering density bound)**.** *Let $L \geq 1$ and let S be a strict (i.e., with distinct moduli) covering of $\mathbb{Z}/L\mathbb{Z}$ by residue classes whose moduli all divide L. Write $M = \{m_i : (a_i, m_i) \in S\}$ for the set of moduli used. Then*

$$\sum_{m \in M} \frac{1}{m} \;\geq\; 1.$$

Proof. Apply Lemma 3 to S: the cardinality of the covered subset is at most $\sum_{(a,m)\in S} L/m$. Since S covers all of $[0, L)$, the covered cardinality is exactly L, so $L \leq \sum_{(a,m)\in S} L/m$. Strictness implies the moduli m_i are pairwise distinct, so $\sum_{(a,m)\in S} L/m = \sum_{m\in M} L/m = L \sum_{m\in M} 1/m$. Dividing by $L > 0$ gives the stated bound.[1] ☐

Corollary 5 (Abstract density obstruction)**.** *Let $L \geq 1$ and let $M \subseteq \mathbb{N}_{\geq 2}$ be a finite set of candidate moduli, each dividing L. If $\sum_{m\in M} 1/m < 1$, then no strict covering of $\mathbb{Z}/L\mathbb{Z}$ by residue classes with moduli in M exists.*

Proof. Were there such a covering, Theorem 4 applied to the subset $M' \subseteq M$ of moduli actually used would yield $\sum_{m\in M'} 1/m \geq 1$. But $M' \subseteq M$ and reciprocals are positive, so $\sum_{m\in M'} 1/m \leq \sum_{m\in M} 1/m < 1$, a contradiction. ☐

3 The numerical bound at $L = 27720$

Let

$$M_{27720} \;:=\; \mathcal{M} \cap \operatorname{div}(27720) \;=\; \{m \in \mathcal{M} : m \mid 27720\}.$$

Since $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$, the divisor lattice of 27720 has 96 elements; enumerating those which equal $p-1$ for some prime $p \geq 5$ yields a list of 34 admissible moduli, ranging from $m = 4$ (witness $p = 5$) to $m = 9240$ (witness $p = 9241$).

Lemma 6. *The set M_{27720} has 34 elements; explicitly,*

$$\begin{aligned} M_{27720} = \{\, & 4,\ 6,\ 10,\ 12,\ 18,\ 22,\ 28,\ 30,\ 36,\ 40,\ 42,\ 60,\ 66,\ 70,\ 72,\ 88,\ 126,\\ & 180,\ 198,\ 210,\ 280,\ 330,\ 396,\ 420,\ 462,\ 616,\ 630,\\ & 660,\ 990,\ 1320,\ 2310,\ 2520,\ 4620,\ 9240\,\}. \end{aligned}$$

Proof. The 96 divisors of 27720 are the products $2^a 3^b 5^c 7^d 11^e$ with $a \in \{0,1,2,3\}$, $b \in \{0,1,2\}$, $c, d, e \in \{0,1\}$. For each such divisor $m \geq 4$ one checks whether $m+1$ is prime, retaining those for which it is. The listed 34 values are precisely those that pass the check. ☐

Lemma 7. $D(27720) \;:=\; \displaystyle\sum_{m\in M_{27720}} \frac{1}{m} \;=\; \frac{953}{990} \;<\; 1.$

Proof. Direct rational computation over the 34 terms of M_{27720}. The common denominator divides $\operatorname{lcm}(M_{27720}) = 27720$; the sum equals $953/990$, which is strictly less than 1. ☐

Theorem 8 (Numerical lower bound on Erdős #273)**.** *There is no strict covering of $\mathbb{Z}$ by residue classes whose moduli all lie in $\mathcal{M}$ and all divide* $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$.[2]

[1] Formally verified in Lean 4 + Mathlib (commit `c5ea00351c28`); see `formal/Erdos273/CoveringSystem.lean`, theorems `strict_covering_density_bound` and `strict_covering_reciprocal_bound`.

[2] Formally verified in Lean 4 + Mathlib (commit `c5ea00351c28`); see `formal/Erdos273/IntegerForm.lean`, theorem `no_strict_admissible_covering_Z_27720`, which combines the finite-range theorem `no_strict_admissible_cover_27720` in `formal/Erdos273/NumericalBound27720.lean` with the $\mathbb{Z}$-to-$[0, L)$ equivalence lemma `covers_int_iff_covers_range_L`.

Proof. Since $27720 \in \mathbb{N}_{\geq 1}$ and every element of M_{27720} satisfies $m \geq 4 \geq 2$ and $m \mid 27720$, the hypotheses of Corollary 5 apply with $L = 27720$ and $M = M_{27720}$. The inequality $D(27720) < 1$ follows from Lemma 7. Hence no strict covering of $\mathbb{Z}/27720\mathbb{Z}$ by residue classes with moduli in M_{27720} exists. To lift from $\mathbb{Z}/27720\mathbb{Z}$ to $\mathbb{Z}$ we use the standard equivalence: a finite family S of residue classes whose moduli all divide L covers $\mathbb{Z}$ if and only if it covers $\{0, 1, \ldots, L-1\}$. The forward direction is trivial; the reverse follows by reducing any $n \in \mathbb{Z}$ to its representative $r := n \bmod L \in [0, L)$, observing that $n \equiv r \pmod{L}$ and therefore $n \equiv r \pmod{m}$ for every modulus $m \mid L$. This equivalence is the lemma `covers_int_iff_covers_range_L` cited in the footnote. $\square$

Corollary 9. *If Erdős Conjecture 1 has a positive answer, then any witnessing strict covering must use at least one modulus $m \in \mathcal{M}$ with $m \nmid 27720$.*

Proof. The contrapositive of Theorem 8: if every modulus divided 27720, the covering would be ruled out. $\square$

The same machinery applies verbatim at any period L whose admissible divisors have reciprocal sum below 1. The smallest such L that is not already subsumed by 27720 is $L = 83160 = 27720 \cdot 3$, obtained by adjoining a third factor of 3.

Theorem 10 (Strengthened numerical bound at $L = 83160$)**.** *There is no strict covering of $\mathbb{Z}$ by residue classes whose moduli all lie in $\mathcal{M}$ and all divide* $83160 = 2^3 \cdot 3^3 \cdot 5 \cdot 7 \cdot 11$.[3]

Proof. The divisor lattice of 83160 has 128 elements, of which 45 are admissible; they form

$$M_{83160} = M_{27720} \cup \{108,\ 270,\ 378,\ 540,\ 756,\ 2376,\ 2970,\ 4158,\ 7560,\ 8316,\ 16632\},$$

the eleven new moduli being precisely the admissible divisors of 83160 divisible by $3^3 = 27$ – those made available by the third factor of 3, which $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$ lacks (the smallest is $108 = 109 - 1$). Their reciprocal sum is

$$D(83160) = \sum_{m \in M_{83160}} \frac{1}{m} = \frac{27241}{27720} \approx 0.9827 < 1,$$

so Corollary 5 applies with $L = 83160$ and $M = M_{83160}$; the $\mathbb{Z}$-to-$[0, L)$ equivalence then lifts the conclusion from $\mathbb{Z}/83160\mathbb{Z}$ to $\mathbb{Z}$ exactly as in the proof of Theorem 8. Since $27720 \mid 83160$, this statement strictly contains Theorem 8: every strict admissible covering ruled out there has all of its moduli dividing 83160 as well, and is therefore ruled out here too. $\square$

Adjoining instead a new prime to the period yields an obstruction that is genuinely independent of the previous two.

Theorem 11 (Non-existence at $L = 180180$)**.** *There is no strict covering of $\mathbb{Z}$ by residue classes whose moduli all lie in $\mathcal{M}$ and all divide* $180180 = 2^2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13$.[4]

Proof. The divisor lattice of 180180 has 144 elements, of which 47 are admissible; their reciprocal sum is

$$D(180180) = \sum_{m \in M_{180180}} \frac{1}{m} = \frac{173003}{180180} \approx 0.9602 < 1,$$

so Corollary 5 applies with $L = 180180$, and the $\mathbb{Z}$-to-$[0, L)$ equivalence lifts the conclusion to $\mathbb{Z}$ as before. Unlike Theorem 10, this bound is *not* a strengthening of Theorem 8: since $27720 \nmid 180180$ (the exponent of 2 drops from 3 to 2), the two obstructions exclude different families of covers. The admissible modulus $52 = 53 - 1$, available at neither 27720 nor 83160, enters here through the new prime factor 13. $\square$

[3] Formally verified in Lean 4 + Mathlib (commit `c5ea00351c28`); see `formal/Erdos273/NumericalBound83160.lean`, theorem `no_strict_admissible_covering_Z_83160`.

[4] Formally verified in Lean 4 + Mathlib (commit `c5ea00351c28`); see `formal/Erdos273/NumericalBound180180.lean`, theorem `no_strict_admissible_covering_Z_180180`.

Remark 12. The non-existence of a strict admissible covering at $L = 27720$ is the largest in a sequence of bounded-period non-existence statements established by SAT search in the script `formal/numerics/sat_covering.py`. The CaDiCaL solver [6] confirms UNSAT at $L \in \{60, 120, 360, 840\}$ directly (each instance has $D(L) < 1$ and is decided by the solver in under four seconds); for $L \in \{2520, 27720\}$ the density argument of Theorem 4 is sharper than CDCL search at the chosen budgets. Continuing along this canonical chain, the next period at which $D(L) \geq 1$ is $L = 360360$, where $D(L) = 1.0237$; the globally smallest period with $D(L) \geq 1$ is in fact $L = 55440$ (see §4). At such L the density argument no longer applies and the bounded-period question becomes genuinely combinatorial.

4 Concluding remarks

Theorem 4 and its abstract corollary 5 are stated and proved without arithmetic content – they apply to any set of admissible moduli $\mathcal{M}$, not just $\{p-1 : p \geq 5 \text{ prime}\}$. The numerical bound at $L = 27720$ (Theorem 8) is one instance; the same machinery can be re-applied at other periods.

Extending the density obstruction beyond $L = 27720$**.** An exact-rational scan of $D(L)$ over candidate periods $L \leq 9 \cdot 10^5$ (script `formal/numerics/density_scan.py`)[5] identifies 27 further values $L > 27720$ at which $D(L) < 1$ holds and at least one admissible modulus of L does not divide 27720 – meaning each gives a non-existence statement strictly stronger than Theorem 8. The smallest such L is $L = 83160 = 2^3 \cdot 3^3 \cdot 5 \cdot 7 \cdot 11$, at which $D(83160) = 27241/27720 \approx 0.9827$; the corresponding non-existence statement is Theorem 10, verified above. The smallest L that introduces a new prime factor is $L = 180180 = 2^2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, at which $D(180180) = 173003/180180 \approx 0.9602$. The full non-existence statement at 180180 is Theorem 11, verified above; because $27720 \nmid 180180$, it is independent of the 27720 and 83160 bounds rather than a strengthening of them. Both larger bounds are obtained by the same route: computing the divisor lattice as a pointwise product of prime-power divisor sets via `Nat.divisors_mul`, then discharging each primality obligation with `norm_num` rather than kernel `decide`. The smallest L at which $D(L) \geq 1$ is $L = 55440 = 27720 \cdot 2$, where $D(55440) = 6429/6160 \approx 1.0437$ – so the 2-adic chain $L = 27720 \cdot 2^k$ exits the density-obstruction regime at $k = 1$, while the 3-adic chain $L = 27720 \cdot 3^k$ stays below 1 at least through $k = 3$ (at $L = 748440$).

Outside the bounded-period regime. The elementary density obstruction, decisive for the periods above, cannot by itself settle Conjecture 1: the reciprocal sum over all of $\mathcal{M}$ is unbounded, so for every threshold there are admissible moduli whose reciprocals already exceed it.

Proposition 13. *The reciprocal sum over the admissible moduli diverges:* $\sum_{m \in \mathcal{M}} 1/m = \infty$.[6]

Proof. Indexing $\mathcal{M}$ by the witnessing prime, the sum is $\sum_{p \geq 5} 1/(p-1)$. For every prime p one has $1/p \leq 1/(p-1)$, so were this series to converge, $\sum_p 1/p$ would converge as well, contradicting the divergence of the prime harmonic series. Restoring or omitting the finitely many terms with $p < 5$ changes the sum by a finite amount, so it diverges over $\mathcal{M}$. $\square$

Consequently, any argument valid for arbitrarily large minimum modulus must come from a structural method rather than from the reciprocal sum alone. The literature offers such methods. Hough's distortion method [2] and its strengthening by Balister–Bollobás–Morris–Sahasrabudhe [3] produce non-existence results for covering systems whose moduli set satisfies a multiplicative-density hypothesis. The signed weighted sum used in their density theorem

[5] Performed in `fractions.Fraction` arithmetic; no floating-point arithmetic enters the verdict.

[6] Formally verified in Lean 4 + Mathlib (commit `c5ea00351c28`); see `formal/Erdos273/DensityDivergence.lean`, theorem `admissible_reciprocal_not_summable`.

likewise fails to converge on $\mathcal{M}$, so that theorem does not apply uniformly here. Their hypothesis-free auxiliary result – Schinzel's conjecture, proved as [3] Theorem 1.2, that any covering contains a pair $m_i \mid m_j$ – is automatically satisfied by $\mathcal{M}$ (which is rich in divisibility pairs) and so yields nothing beyond Proposition 13. One might hope to sharpen the universal bound by exploiting the arithmetic of $\mathcal{M}$ inside the Cummings–Filaseta–Trifonov refinement [4], whose squarefree-moduli bound of ≤ 118 is, mechanically, BBMST's distortion theorem instantiated on a hyperplane reformulation. A numerical reconstruction of that optimisation reproduces the published squarefree value to eight significant figures, confirming the engine; but it also reveals an intrinsic obstruction to specialising it to $\mathcal{M}$. The natural $\mathcal{M}$-input is the Dirichlet density $1/(p-1)$ of primes q with $p \mid q-1$. This density measures sparsity *among the primes* q, whereas the distortion bound counts the prime-support *patterns* that the moduli realise – and every finite pattern $\{p_1, \ldots, p_r\}$ is realised by some $m = q-1 \in \mathcal{M}$, since $q \equiv 1 \pmod{p_1 \cdots p_r}$ has prime solutions by Dirichlet's theorem. The sparsity is therefore invisible to the count: inserting the $1/(p-1)$ factor (whether as a per-coordinate weight or on the hyperplane count, which one checks are algebraically identical) discounts a saving the covering never pays, and over-reaches to the false conclusion that $\mathcal{M}$ cannot cover $\mathbb{Z}$ at all. Consequently the distortion method yields for $\mathcal{M}$ only the bound it gives for arbitrary distinct moduli, with no improvement from the $p-1$ structure. A genuinely sharper non-existence result, if one exists, must come from a mechanism that sees more than the prime-support patterns of the moduli.

A Formal verification

The mathematical content of Sections 2 and 3 has been formally verified in Lean 4 against the Mathlib library (commit `c5ea00351c28`). The proof scripts live in this manuscript's repository under `formal/` and compile under `lake build`. For every verified statement listed below, `#print axioms` reports a dependency chain consisting of only the three Lean foundational axioms {`propext`, `Classical.choice`, `Quot.sound`}. In particular, no `sorry`, no project-local `axiom`, and no `native_decide` (which would inject the axiom `Lean.ofReduceBool` into the dependency chain) is used.

All declarations live in the Lean namespace `Erdos273`; identifiers below are referenced without the namespace prefix to keep the lines readable.

- Lemma 2 (`card_covered_eq`),
- Lemma 3 (`density_union_bound`),
- Theorem 4 (`strict_covering_density_bound`; rational form `strict_covering_reciprocal_bound`),
- Corollary 5 (`no_strict_cover_of_low_density`):

all in `formal/Erdos273/CoveringSystem.lean`.

- Lemma 6: enumeration (`M27720_eq_filter_divisors`) and cardinality (`M27720_card`),
- Lemma 7 (`D27720_lt_one`),
- Finite-range form of Theorem 8 (`no_strict_admissible_cover_27720`):

all in `formal/Erdos273/NumericalBound27720.lean`;

- the $\mathbb{Z}$-to-$[0, L)$ equivalence (`covers_int_iff_covers_range_L`),
- Theorem 8, $\mathbb{Z}$-form (`no_strict_admissible_covering_Z_27720`),

- Corollary 9 (`erdos273_modulus_not_dividing_27720`):

all in `formal/Erdos273/IntegerForm.lean`;

- the divisor-lattice identity `Nat.divisors 83160` $= D_{83160}$ (`divisors_83160_eq`), via `Nat.divisors_mul`,
- $D(83160) = 27241/27720 < 1$ (`D83160_lt_one`) and the admissibility of M_{83160} (`M83160_admissible`),
- Theorem 10, both the finite-range form (`no_strict_admissible_cover_83160`) and the $\mathbb{Z}$-form (`no_strict_admissible_covering_Z_83160`):

all in `formal/Erdos273/NumericalBound83160.lean`;

- the divisor-lattice identity `Nat.divisors 180180` $= D_{180180}$ (`divisors_180180_eq`),
- $D(180180) = 173003/180180 < 1$ (`D180180_lt_one`) and the admissibility of M_{180180} (`M180180_admissible`),
- Theorem 11, finite-range (`no_strict_admissible_cover_180180`) and $\mathbb{Z}$-form (`no_strict_admissible_covering_Z_180180`):

all in `formal/Erdos273/NumericalBound180180.lean`. Finally, Proposition 13 (`admissible_reciprocal_not_summable`) is in `formal/Erdos273/DensityDivergence.lean`.

The formalisation defines a `ResidueClass` structure recording the modulus $m \geq 2$ and the residue (canonically as an element of `ZMod m`); a `CoveringSystem` structure packaging a `Finset ResidueClass` with the covering proof; and a predicate `CoveringSystem.IsStrict` encoding the distinctness of moduli. The set $\mathcal{M}$ is implemented as `Erdos273.AdmissibleModulus`, a decidable predicate; the set M_{27720} is implemented as an explicit `Finset` of 34 natural numbers and proved equal to `Nat.divisors 27720` filtered by admissibility (kernel `decide`, not `native_decide`).

Headline statements (Lean source)

The statements below are quoted verbatim from the Lean sources; all declarations live in `namespace Erdos273`. The core density obstruction, in `formal/Erdos273/CoveringSystem.lean`, is

```
theorem strict_covering_density_bound
    (S : Finset ResidueClass) (L : ℕ)
    (_hStrict : (S.image ResidueClass.modulus).card = S.card)
    (hdvd : ∀ c ∈ S, c.modulus ∣ L)
    (hcov : ∀ i : ℕ, i < L → ∃ c ∈ S, (i : ℤ) ∈ c.toSet) :
    L ≤ ∑ c ∈ S, L / c.modulus
```

The partial result on Conjecture 1 itself, in $\mathbb{Z}$-form (`formal/Erdos273/IntegerForm.lean`), reads

```
theorem erdos273_modulus_not_dividing_27720 :
    Erdos273Conjecture → ∃ C : CoveringSystem, C.IsStrict ∧
        C.SolvesErdos273 ∧ ∃ c ∈ C.classes, ¬ c.modulus ∣ 27720
```

and the divergence of the reciprocal sum over $\mathcal{M}$ (`formal/Erdos273/DensityDivergence.lean`) is

```
theorem admissible_reciprocal_not_summable :
    ¬ Summable (fun p : Nat.Primes ↦ (1 : ℝ) / ((p : ℝ) - 1))
```

Axiom witness

For each headline theorem, `#print axioms` reports exactly the three Lean foundational axioms and nothing else:

Theorem	File	Axioms
`strict_covering_density_bound`	`CoveringSystem.lean`	clean
`no_strict_cover_of_low_density`	`CoveringSystem.lean`	clean
`erdos273_modulus_not_dividing_27720`	`IntegerForm.lean`	clean
`no_strict_admissible_covering_Z_83160`	`NumericalBound83160.lean`	clean
`no_strict_admissible_covering_Z_180180`	`NumericalBound180180.lean`	clean
`admissible_reciprocal_not_summable`	`DensityDivergence.lean`	clean

Here "clean" abbreviates the axiom set {`propext`, `Classical.choice`, `Quot.sound`}, and file paths omit the common prefix `formal/Erdos273/`.

Reproducing the verification

1. Clone the repository at the commit whose `formal/lake-manifest.json` pins Mathlib at `c5ea00351c28`.
2. Build with `(cd formal && lake exe cache get && lake build)`: the blanket `import Mathlib` pulls the pre-built `.olean` cache, after which the project's own modules compile in a few minutes.
3. Inspect the axiom report: every `.lean` file ends with `#print axioms` lines for its theorems, emitted during `lake build`; the output lists only `propext`, `Classical.choice`, and `Quot.sound`.

B Bounded-period SAT enumeration

Independent of the density argument, the script `formal/numerics/sat_covering.py` encodes the bounded-period strict covering problem at user-specified L as a propositional CNF (one Boolean variable per candidate pair (a, m) with $m \in M_L$, $0 \leq a < m$, distinctness as at-most-one constraints per modulus, covering as "at least one variable true" constraints per residue $i \in [0, L)$). The script invokes `cadical` [6] via DIMACS subprocess. The empirical results recorded above (Remark 12) are reproducible from the script.

The SAT outputs are not, on their own, machine-checked: certifying a CDCL solver's UNSAT verdict in Lean would require importing an LRAT-checker, which is not yet in this project. The density argument (Theorem 4) supersedes the SAT certificates for every UNSAT verdict in the canonical sequence $L \in \{60, 120, 360, 840, 2520, 27720\}$, so the formally-verified statement (Theorem 8) does not depend on the SAT solver. The SAT enumeration remains useful as a check for $L \geq 360360$, where the density argument no longer applies and the question becomes genuinely combinatorial.

References

[1] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, L'Enseignement Mathématique **28** (1980), Problem #273, p. 24.

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